State of the Market Address

and other post-graduation stuff

General Market Trends

From the State Board

Figure 2. Estimate of Washington State Computer and Mathematical Sciences (SOC 15-0000) Employees from 2017 to 2024

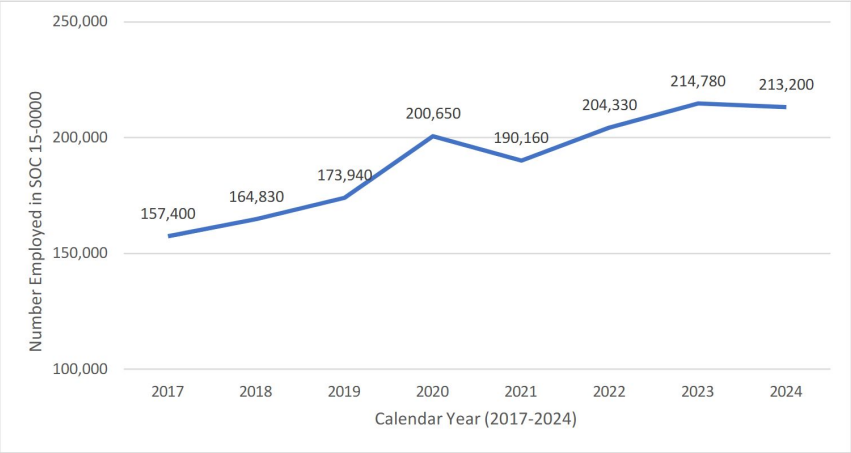
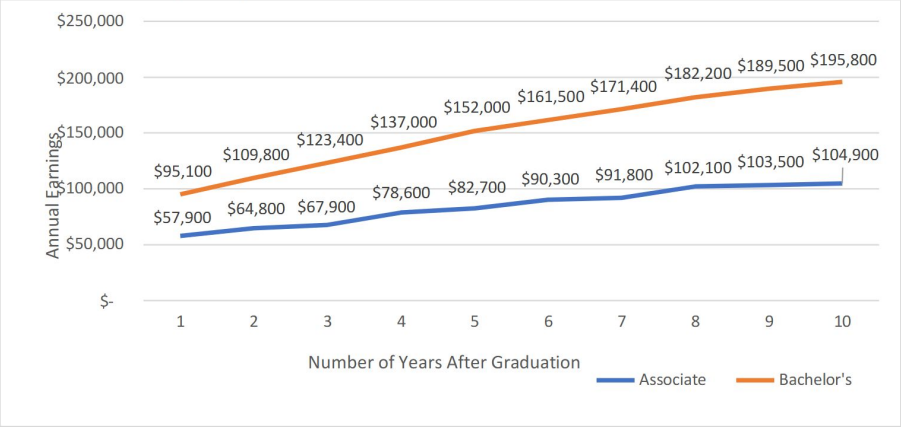


Figure 3. Inflation-Adjusted Median Annual Earnings for Washington Computer Science and Other Tech (CIP 11) Graduates from 1 to 10 Years After Graduation



Source: Data for this figure were drawn from the Washington EDRC dashboard on January 18, 2026.

General Market Trends

From the State Board

- The market is more competitive, especially at the entry level
- State-level data still show long-term growth in tech employment
- AI is reshaping work, but not eliminating the need for software developers
- Internships and practical experience matter more than ever

Entry Level Tech Jobs

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5425555

IMO it's the market more than AI.

Junior hiring has been getting shakier for years.

AI doesn't help. It also isn't the main cause.

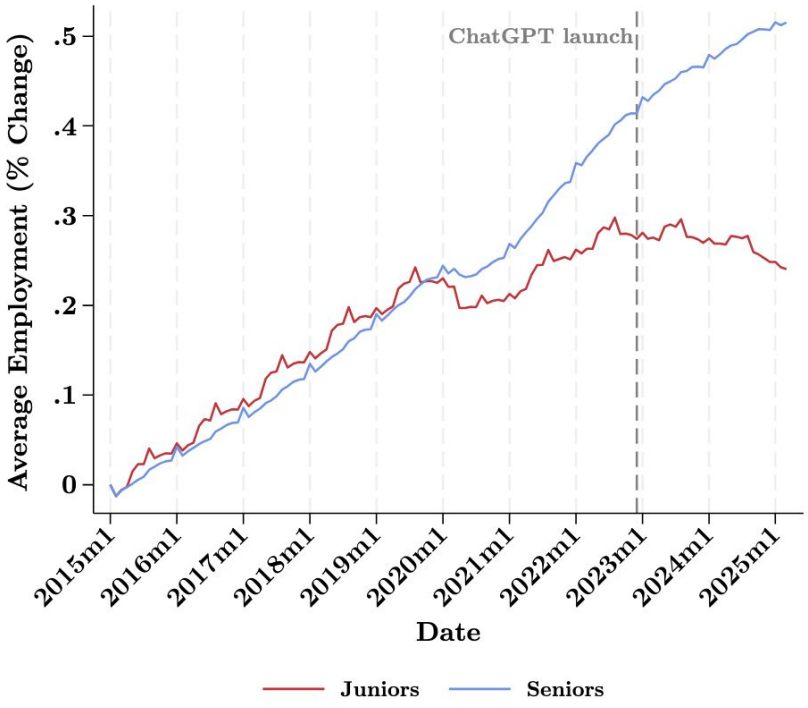


Figure 1: Log Average Employment of Juniors and Seniors in Sample Firms

High Level Takeaway

It’s not doomed. Nor is it smooth sailing. The landscape is changing. Opportunities exist, but are in flux.

Tech has always been an extremely fast moving field with boom-bust cycles and disruption. It comes with the territory.

This kind of environment tends to require flexibility and initiative.

Practical Advice

So you’re about to graduate and still want to work in tech. What do you do?

- Don’t filter yourself
- Looking beyond SWE
- What if I don’t get a job?
- Certs & degrees

Am I Qualified

What Do You Have Experience With?

What Do You Have Experience With?

- Software and Web Development Experience
  - Full stack development with HTML, CSS, JS, PHP, Python, Java, SQL
  - Software practices like version control, testing, debugging
- Data Experience
  - Handling, storing, and managing data
  - Applying data and AI knowledge to real projects
- Deployment Experience
  - Deployment processes
  - Cloud services
- Project / Team Management Experience
  - Teamwork and technical collaboration
  - Agile methodology
  - Technical project management

Transferable Skills

- Problem Solving, debugging, troubleshooting
- Communication and teamwork
- Fast learning and adaptability to new tool and tech
- Data and AI literacy
- Project and time management

Many roles require these skill sets!

Am I Qualified?

How do you know if you’re qualified for a particular position?

Am I Qualified?

How do you know if you're qualified for a particular position?

You often won't know just from a job posting. So don't rule yourself out!

Am I Qualified?

You Don’t Need to Match 100% of the Requirements

- If you meet the majority of the qualifications, apply!
- Employers write wish lists, not strict requirements

Example: If the post lists "Experience in JavaScript, React, and Python" and you’ve used two out of three, that’s a strong match.

If you’ve built something similar (even in school projects), it counts as experience.

Am I Qualified?

Don’t Worry About 2-3 Years of Experience

- Many “2-3 years of experience” roles are actually open to recent grads
- If you can demonstrate equivalent project experience, you’re qualified

Apply If

- You understand most of the tools or can learn them quickly
- You’ve done something similar in a class, project, or internship

Example: A job lists “AWS experience,” but you’ve deployed apps in another cloud environment. That’s usually close enough.

When You Might Be Underqualified

How do you know if you’re underqualified for a particular position? What are the red flags?

When You Might Be Underqualified

“Must Have” Requirements You Can’t Meet. Job postings that explicitly state non-negotiable skills, certifications, or experiences.

Examples:

- “Must have AWS Solutions Architect Certification.”
- “Required: 5+ years experience managing production Kubernetes clusters.”
- “Extensive experience optimizing large-scale distributed systems with low-latency requirements.”

When You Might Be Underqualified

Technical Leadership or Management Responsibilities.

Example: “Oversee and mentor junior engineers while leading product development cycles.”

Industry-Specific Experience You Lack. Some roles require deep knowledge of a specific industry or domain.

Example: “10+ years of experience in the healthcare IT industry.”

Tech Adjacent Jobs

Being a software developer is one of many options

Specialize Your Search

If you search things like:

- “Software Developer Jobs”
- “Full Stack Developer Jobs”
- “Remote Developer Jobs”

Then you are competing with the highest possible number of people.

Refine and specialize your search to reduce competition.

What Else?

What other technical jobs or roles could you potentially fill?

Software / Web / Application Development

- Software Developer/Engineer
- Application Developer
- Mobile Developer
- Platform Engineer
- Front/Back/Full Stack Developer
- Game Developer
- Embedded Software Developer
- Content Management System (CMS) Developer
- (Service) Cloud Engineer
  - Azure, AWS, GCP, etc.
- Systems / Software Architect
- (Tool/Language) Developer
  - Java, JavaScript, C#, .NET, etc.
- Web Developer / Designer
- User Interface (UI) Developer / Designer
- User Experience (UX) Developer / Designer

DevOps

- DevOps Engineer
- Site Reliability Engineer
- Infrastructure Engineer
- CI/CD Engineer
- Configuration Management Engineer
- Release Manager

Data Science / Analytics / Visualization and AI

- Data Analyst
- Data Scientist
- Data / Big Data Engineer
- Business Analyst
- Machine Learning (ML) Engineer
- Artificial Intelligence (AI) Engineer
- Business Intelligence (BI) Developer / Engineer / Analyst
- IoT Analytics Specialist
- Data Visualization Designer
- Data Architect

IT and Networking

- Systems Administrator
- Database Administrator
- Network Engineer
- IT Specialist
- Cloud Administrator
- Cyber Security Analyst
- Information Security Analyst
- Compliance Analyst

Project Management and Consulting

- SCRUM Master
- Product Owner
- Project Manager
- Product Manager
- Agile Coach
- Technical Project Manager
- IT Consultant
- Technical Consultant
- Software Consultant
- Technology Analyst
- Business Analyst

Low-Code Development and Integration

- Low-Code Developer / Engineer
- Low-Code Platform Architect
- Power Platform Developer
- Integration Specialist
- Salesforce Developer
- Dynamics 365 Developer

Quality Assurance and Testing

- Quality Assurance (QA) Analyst / Engineer / Specialist
- Software Tester
- Software Development Engineer in Test (SDET)
- Automation Test Engineer
- QA Automation Engineer

Support and Training

- Application Support Engineer
- Technical Support Analyst / Engineer
- Customer Success Engineer
- IT Support Specialist
- Help Desk Technician
- Software / Application Trainer
- Implementation and Training Specialist

And there are many more roles in areas like…

- Research and academia
- Specialized roles (AR/VR, Blockchain, Robotics, IoT)
- Freelance development and analytics
- Cyber and digital security
- Healthcare and wellness
- Education
- And **LOTS** more… technology is a part of everything

If you are willing to be flexible, creative, and take the initiative, opportunities exist.

What If I Still Can’t Find A Job?

Tech-ify the Non-Tech: Most jobs involve tech, even if they aren’t software development (project coordinator, analyst, admin, etc). It is applicable to almost everything. The skillset is valuable and highly transferable. Automate spreadsheets with Python or build an internal dashboard for your office. These are valuable real-world uses of your skills.

Contracting and Gig Work: Even a $200 contract allows you to list "Freelance Developer" on your resume with a real client name and can make connections.

What If I STILL Can’t Find A Job?

If you’re still interested in tech, I’d suggest the following:

- Don’t quit! Keep building.
- DO NOT build the next big thing. Build **REAL** tools you can **DEPLOY** to actual **USERS** within **WEEKS**. Even if the scope is small.
  - Local business, non-profits, communities, etc.
- Stay in the community. Network! Go to events and meetups. Reach out.
- Find open-source communities and contribute to software. Find a tools you use or like. Fix bugs, write documentation, or add features. Talk to people!

Don’t disengage or sidetrack into multi-month solo projects. Your skills atrophie.

A Note on Networking

Networking is not optional in tech and it never has been. Even when I was applying to tech roles between 2019 and 2023:

- I got my first part-time jobs teaching K-12 STEM from talking to administrators
- I got my first internship from talking to recruiters at an event at GRC
- I got my first job from making a good impression during the internship
- I got my first project by reaching out to senior people and making connections
- My second project found me. Total luck
- I got my third project came from networking with the talent lead for our region
- I got my job here because I knew Tina and reached out about helping at GRC
- I got my associate faculty role because Ken told me it existed and to apply

What About Master’s / Certs?

Let’s talk about those.

Continuing Education: Certifications & Advanced Degrees

For Software Developers / Tech Workers

Why Continue Your Education?

- Stay competitive in a changing field
- Fill knowledge gaps (e.g., cloud, security, AI)
- Unlock new job roles or promotions
- Build credibility and confidence

Certifications: What Are They?

- Proof of specific technical skills
- Often vendor-specific (e.g., AWS, Microsoft)
- Usually involve:
  - Study & practice
  - Passing an exam (sometimes multiple levels)
- Typically costs $100 - $200 to take the exam

Popular Certifications

Cloud: AWS Cloud Practitioner, Azure Fundamentals, AWS Solutions Architect, Certified Kubernetes Admin

Security: CompTIA Security+, CISSP

Development: Oracle Java Certification, Azure Developer Associate

Project/Team Management: Certified ScrumMaster (CSM), Professional Scrum Master (PSM)

Do Certifications Matter?

✅ **Certs:**

- Can boost your resume
- Show commitment & specialized skills

⚠️ **But:**

- Don't replace real-world experience
- Hiring managers typically value projects, internships, & teamwork more

How Much Do Certs Help?

Certification	Dev Roles	Specialized Roles
Cloud (AWS, Azure)	Can be helpful	May be expected in advanced cloud roles
Security (Security+, CISSP)	Nice-to-have	Essential in certain security careers
Development (Java, Azure Dev)	Limited impact	Could be important for platform-specific roles
Project Mgmt (CSM, PSM)	Minimal for devs	Helpful for Scrum Master/Product roles

Pursuing a Master's Degree

- **Masters of Science in CS / Data / AI / Cybersecurity**
- Time: 1–3 years (part/full-time)
- Often online or flexible options available

Why?

- Deeper specialization
- Career pivot (e.g., move into AI, research, teaching)
- Helpful for roles in: R&D, academia, and some tech firms

When Is a Master's Degree Required?

✅ **Commonly Needed For:** Machine Learning Engineer, Data Scientist / AI Specialist, Research & Development roles, Advanced Cybersecurity (e.g., cryptography), Academic / Teaching positions

⚠️ **Rarely Required For:** Full-stack / Frontend / Backend Developer, Mobile & Web Development, DevOps / Site Reliability Engineer, QA / Test Automation roles

If you're aiming for a **highly specialized or research-driven role,** a Master’s (or higher) can be a gateway. For **most dev roles, real-world experience matters most.**

When is a Master's Worth It?

✅ Good fit if:

- It's required for your field
- You want to pivot into a new field
- You aim for leadership / R&D roles
- Your employer offers tuition support

⚠️ But maybe not if you want to stay in coding/dev roles or are mainly seeking a salary bump.

Master's Degree vs Certifications

Aspect	Master's Degree	Certifications
Time Commitment	High (1–3 years)	Low–Medium (weeks–months)
Cost	High ($15k–$60k+)	Low ($100–$1,500)
Depth of Knowledge	Broad, Deep, Often Theoretical	Narrow, Practical
Career Impact	Long-term Credibility	Short-term Skill Proof

When to Pursue a Master's?

- After 2-5 years of work experience
- When you have clarity on your career goals
- If you plan to specialize or shift focus

Talk to alumni and faculty before committing.

DO NOT apply without first speaking directly to at least one person in the program!

	Online	In-Person
Flexibility	✅ High	⚠️ Low
Cost	✅ Often lower overall	⚠️ Usually much higher
Learning	Self-paced options	Structured, immersive
Networking	⚠️ Limited (virtual events/forums)	✅ Strong (face-to-face connections)
Resources	⚠️ May be limited	✅ Full campus access
Perception	⚠️ Credible if reputable	✅ Still the gold standard

Typically, top employers care more about program quality and experience than format. Choose accredited, recognized schools.

Key Takeaways

- Certifications: Great for quick skill boosts & marketability
- Master's Degree: Strategic, long-term investment

Neither replaces real-world experience. Combine learning with doing!

Questions?

What About Master’s / Certs?

I would suggest not pursuing a master’s degree to avoid or escape the job search.

- They are expensive, especially in-person
- They can help your resume, but don’t guarantee a job
- Most of you need experience more than additional schooling
- There’s no rush, you can do them later

If you have a specific goal or clear plan go for it. Just make sure it is a proactive choice to set you up for something clear and specific.

The most valuable thing about a master’s is networking and internships!